Search Smooths Discontinuities: Updates

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Previous story: Goal

Goal: show that directed search can smooth discontinuities in DC-EGM problems
— as an alternative to EV Type I taste shocks

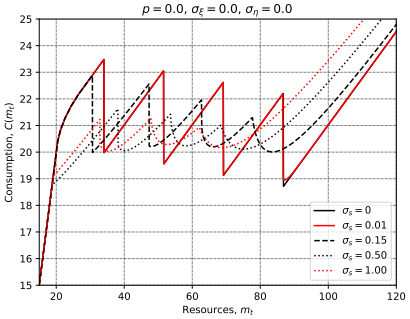
- Standard fix (Iskhakov et al. 2017): EV taste shocks randomize the discrete outcome
- Search: agents choose effort $\varepsilon_t \in [0, 1]$ that raises the probability of switching
- Smoothness is part of the agent's economic problem, not only a numerical trick

Application: consumption–saving retirement model (work vs retire)

Previous story: search alone is not enough

- Search smooths only to the *right* of discontinuities
- Low-M workers optimally set $\varepsilon^* = 0$ (plan to keep working)

$\Rightarrow$ Asymmetric / one-sided smoothness



$p = 0$: smooths right of kinks

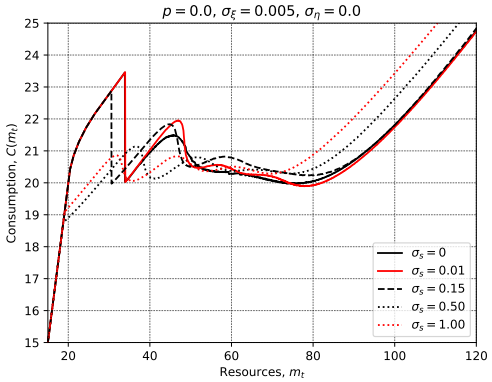
Previous story: candidate complements to search

To get uniform smoothness around discontinuities, combine search with:

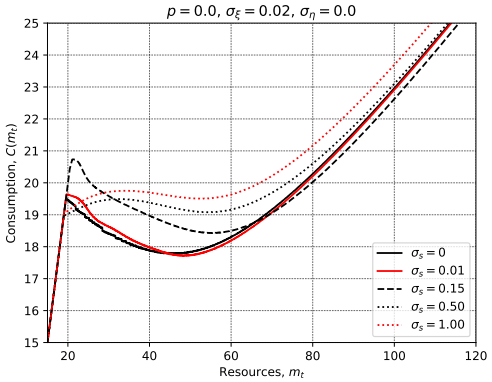
1. **Taste shocks** σ_ξ (multiplicative log-normal on the value of working no EV)
2. **Exogenous option probability** p (probability of receiving the option without effort)
3. **Income shocks** σ_η (transitory income risk)

All three appear in the retirement application of the paper.

Previous story: search + taste (+ income) delivers smoothness



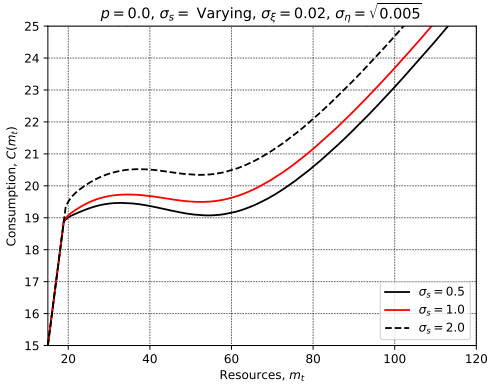
Small taste: $\sigma_\xi = 0.005$



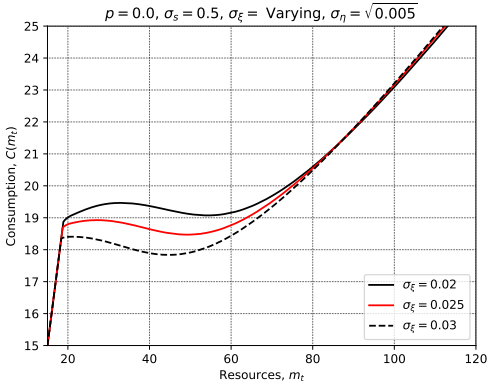
Larger taste: $\sigma_\xi = 0.02$

- Taste expands the region where agents search around kinks
- With moderate σ_s and σ_ξ, discontinuities are largely smoothed

Key insight: search and taste shift c in opposite directions



Raising σ_s shifts c **up**



Raising σ_ξ shifts c **down**

- Search: utility cost of search $\Rightarrow$ counterbalance with higher c
- Larger taste shocks: risk of early retirement: $\Rightarrow$ higher savings $\Rightarrow$ lower c

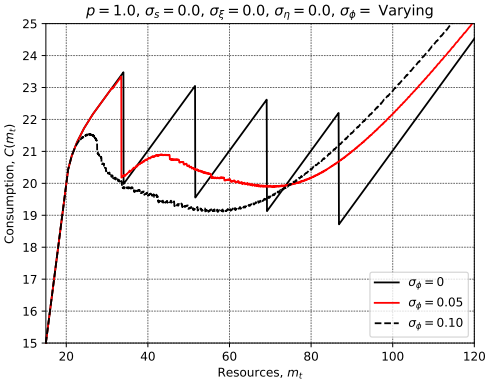
Update 1: introduce price / account dispersion

Idea: contingent on receiving the option, draw a rejectable realization and then accept or deny

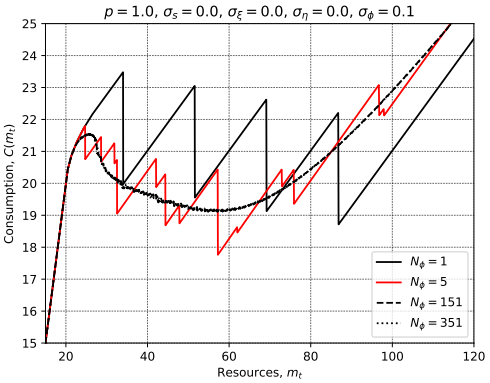
- Retirement: stochastic account factor ϕ_t with $\tilde{M}_t = \phi_t M_t$
- Durables: stochastic purchase price p_t on new durable units
- Low draw accepted, high draw rejected $\Rightarrow$ softens the extensive margin

$$V_t(M_t, \phi_t) = \max\left\{ V_t^{\mathrm{Ret}}(\phi_t M_t),\ V_t^{W}(M_t) \right\}, \quad \bar{V}_t = \mathbb{E}_\phi[V_t]$$

Update 1: smoothing through price dispersion



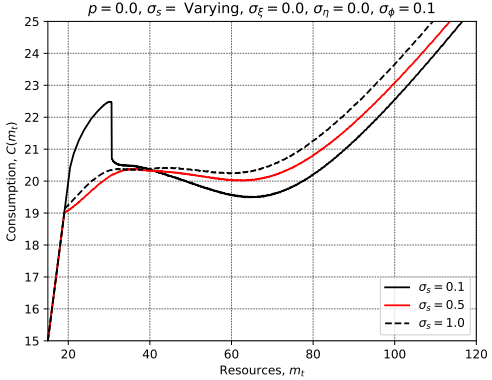
Larger σ_ϕ smooths more

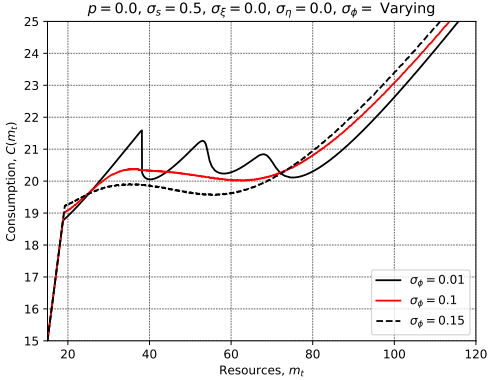


Need enough ϕ-nodes

Key insight: search + price dispersion deliver smoothness

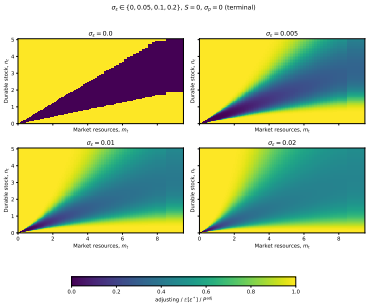
- Search governs *whether* a draw arrives; ϕ_t / p_t is the *content* of the draw
- Complements: option value of the draw raises search effort



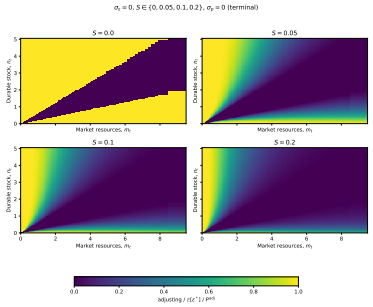


Update 2: durable keep–vs–adjust model in T

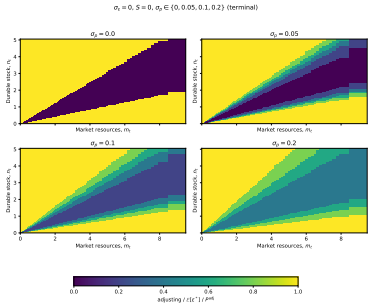
- Discrete choice: keep $d_t = n_t$ vs adjust at cost τ with liquidity $\ell_t = m_t + (1-\tau)n_t$
- Three smoothers: EV taste σ_ε, directed search S, price dispersion σ_p
- Hard max $\Rightarrow$ S,s bands and discontinuous policies; smoothers soften the band



Taste



Search



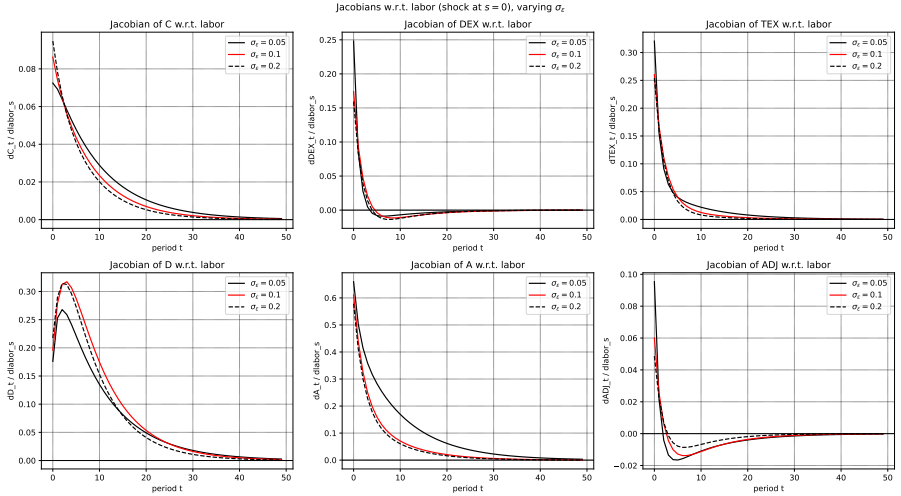
Prices

Key insight: all three smooth; SS effects diverge

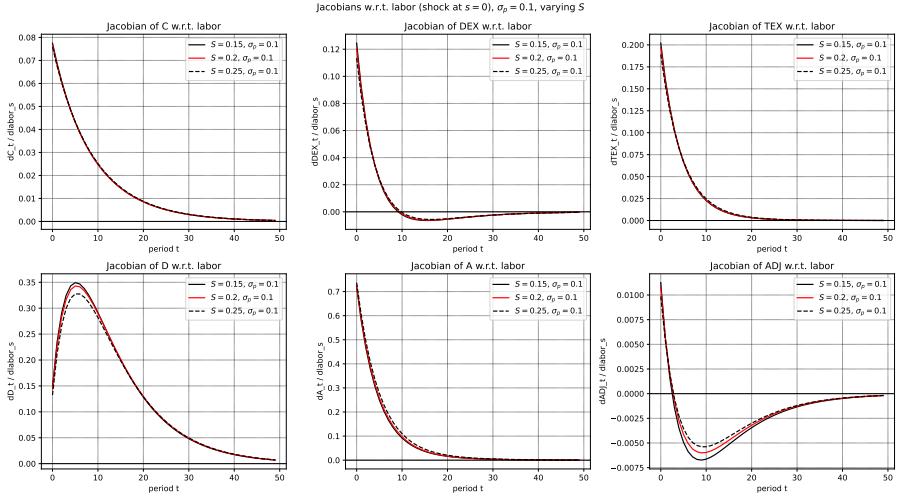
		Steady state					
		C	D	DEX	TEX	A	ADJ (%)
σ_ε	0.05	0.82	3.38	0.17	0.99	1.74	9.8
	0.1	0.80	3.31	0.17	0.97	1.25	13.5
	0.2	0.79	2.97	0.15	0.94	0.66	25.0
σ_p $(S = 0.2)$	0.08	0.83	4.35	0.22	1.05	1.60	9.8
	0.1	0.83	4.87	0.24	1.07	1.35	11.6
	0.12	0.83	5.64	0.28	1.11	1.18	13.8
S $(\sigma_p = 0.1)$	0.15	0.83	4.98	0.25	1.08	1.28	12.4
	0.2	0.83	4.87	0.24	1.07	1.35	11.6
	0.25	0.83	4.83	0.24	1.07	1.47	10.8

- Taste ↑: More people exogeneously adjust: Adjusting probability ↑, D ↓, A ↓
- Price dispersion ↑: Better potential deals: Adjusting probability ↑, D ↑, A ↓
- Search ↑: Adjusting becomes more costly: Adjusting probability ↓, D ↓, A ↑

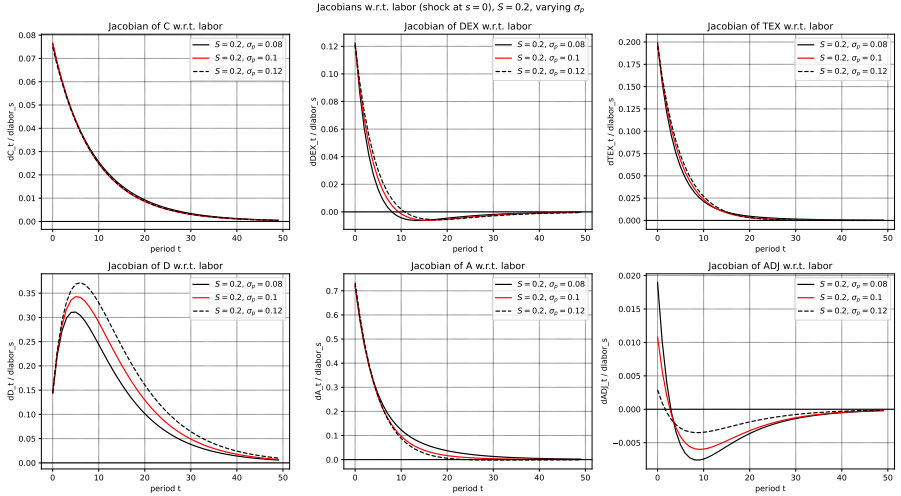
Jacobians: taste changes iMPC impact *and* shape



Jacobians: search changes shape, MPC stays the same



Jacobians: prices change shape, MPC stays the same



Open question: what is the main focus?

1. **Search + price as alternative to EV taste shocks**
   - Need to compare with EV taste shock (not lognormal one)
   - Drop exogenous switch option in retirement problem

   $\Longrightarrow$ Narrow and targeted focus with clear punchline

2. **Compare all smoothing mechanisms side by side**
   - Add exogeneous switch probability and log-normal taste shocks in durable model

   $\Longrightarrow$ Broader focus and more systematic review

Open question: Add mathematical proofs

- Prove that search / prices / taste remove EGM non-monotonicities
- work with Akshay on formal results (cannot fact-check Claude by myself)

Other: To-dos

- Comparison metric: no of kinks, euler errors, convergence no of iterations, solution time
- Estimation/calibration example of why opposite economic effects matter? Similar to Beraja and Zorzi
- Micro data for duration ($\implies$ search cost) and price dispersion